Draft version March 13, 2026
Typeset using LaTeX **twocolumn** style in AASTeX7.0.1

What GW250114 Teaches Us About General Relativity

Jerry Yeung[1]

[1] *Department of Physics and Astronomy, Stony Brook University, Stony Brook, NY 11794-3800, USA*

ABSTRACT

On January 14, 2025, the LIGO detectors observed GW250114_082203, the most powerful gravitational-wave signal recorded to date, with a network signal-to-noise ratio of 76. This binary black hole merger, originating from two black holes of masses $\sim 33.6\,M_\odot$ and $\sim 32.2\,M_\odot$, offers an opportunity for testing Einstein's theory of general relativity in the strong-field, high-velocity regime. In this overview, I will synthesize the key results from analyses of GW250114 and their implications for astrophysics. The event enables multiple independent tests across all phases of the coalescence, black hole spectroscopy reveals at least two quasinormal modes (the fundamental quadrupolar mode 220 and its first overtone 221) with a spectroscopic pattern consistent with the Kerr metric prediction, while the hexadecapolar mode (440) is constrained for the first time in a single event. Full inspiral-merger-ringdown analyses yield constraints on post-Newtonian parameters that are 2–3 times more stringent than those obtained by hierarchically combining dozens of events from the fourth Gravitational-Wave Transient Catalog (GWTC-4.0). Signal consistency tests confirm the Hawking area theorem at 4.8σ credibility and find no excess residual power. This paper serves as an overview and the potential implications from Abac et al. (2025).

1. INTRODUCTION

On January 14, 2025, the LIGO detectors (Collaboration et al. 2015) recorded the most powerful gravitational-wave signal observed to date: GW250114_082203 (hereafter GW250114) (Abac et al. 2025). With a network signal-to-noise ratio of 76, this binary black hole merger represents an unprecedented opportunity for testing Einstein's theory of general relativity (Einstein 1915) in the extreme gravity regime. The signal's exceptional loudness, approximately three times that of the first detection GW150914 (Abbott et al. 2016), stems from the merger of two massive black holes with component masses of $33.6^{+1.2}_{-0.8}\,M_\odot$ and $32.2^{+0.8}_{-1.3}\,M_\odot$, dimensionless spin magnitudes ≤ 0.24 and ≤ 0.26, and eccentricity constrained to $e \leq 0.03$ at a reference frequency of 13.33 Hz (Abac et al. 2025).

General relativity has withstood over a century of experimental scrutiny, from Solar System tests (Will 2014) to binary pulsar observations (Kramer et al. 2021). However, the nature of singularities inside black hole event horizons (Hawking & Penrose 1970) is still unknown, Hawking's information loss paradox (Almheiri et al. 2021; Raju 2022), and the need for dark matter and dark energy to explain cosmological observations (Clifton et al. 2012). These gaps motivate continued searches for possible deviations from GR, particularly in the strong-field, high-velocity regime where alternative gravity theories predict the most pronounced departures from Einstein's predictions (Bagui et al. 2023). Some modified gravity theories admit the Kerr metric as a solution (Kerr 1963), making tests of GR and tests of Kerr spacetime complementary endeavors.

Black holes in vacuum general relativity are uniquely characterized by their masses and spins, the infamous "no-hair theorem" (Israel 1967). When perturbed, these objects emit a characteristic spectrum of damped sinusoids known as quasinormal modes (QNMs) (Press 1971), whose complex frequencies depend solely on the black hole's intrinsic properties. This spectroscopic signature provides a direct observational test of the Kerr metric: a measurement of a single mode suffices to constrain the remnant's mass and spin, while measurements of multiple modes enable a consistency check of the no-hair theorem through the agreement of their properties with the Kerr prediction (Johnson 1981; Visser & Wiltshire 2004)

As Detweiler noted in 1980, detecting black hole QNMs could confirm their existence with a certainty comparable to the way the 21 cm line unequivocally identifies interstellar hydrogen, this launched the black hole spectroscopy program (Detweiler 1980; Gossan et al. 2012).

2. GW250114

GW250114 was detected on January 14, 2025, by the two LIGO interferometers at Hanford and Livingston.

The Virgo and KAGRA detectors were offline during this period, yet the network signal-to-noise ratio reached an exceptional value of 76 (Abac et al. 2025) (also see Fig. 1). This makes GW250114 the loudest gravitational-wave signal observed to date, surpassing even GW150914 by approximately a factor of three in SNR.

Parameter estimation using the NRSur7dq4 waveform model (Varma et al. 2019) revealed that the source was a binary black hole merger at a luminosity distance of approximately 540 Mpc (redshift $z \sim 0.12$) (Abac et al. 2025). The component masses were found to be $33.6^{+1.2}_{-0.8}\,M_\odot$ and $32.2^{+0.8}_{-1.3}\,M_\odot$ (Abac et al. 2025), making this system remarkably similar in mass ratio to GW150914 but significantly closer. The dimensionless spin magnitudes were constrained to ≤ 0.24 and ≤ 0.26 at 90% credibility, indicating relatively slowly spinning black holes (Abac et al. 2025). The orbital eccentricity was constrained to $e \leq 0.03$ at a reference frequency of 13.33 Hz, consistent with the near-circular orbits expected from isolated binary evolution (Abac et al. 2025).

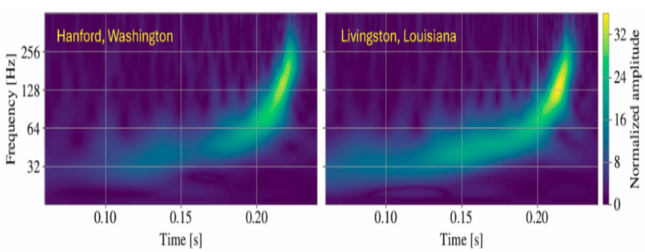

Figure 1. Time-frequency spectrograms of GW250114 as observed by the LIGO Hanford (left) and LIGO Livingston (right) detectors. The signal sweeps from approximately 30 Hz to 250 Hz over two-tenths of a second. The exceptional SNR of 76 is evident from the bright, clear track. Image from LIGO, Virgo and KAGRA (LVK) collaborations.

The extraordinary SNR of GW250114 arises from a fortuitous combination of factors. First, the source masses place the signal peak in the most sensitive frequency band of the Advanced LIGO detectors, around 100-200 Hz. Second, the binary was nearly optimally oriented, with an inclination angle of $0.78^{+0.19}_{-0.23}$ rad, maximizing the observed amplitude (Abac et al. 2025). Third, the source distance was relatively modest by cosmological standards. The result is a waveform with not only exceptional amplitude but also significant contribution from higher-order multipoles, the (4,4) mode alone contributes an SNR of $3.6^{+1.4}_{-1.5}$ to the full signal (Abac et al. 2025).

This event effectively condenses the statistical power of dozens of typical events into a single observation, enabling tests of general relativity that would otherwise require hierarchical combination of the entire catalog.

3. BLACK HOLE SPECTROSCOPY

3.1. *Quasinormal Modes and the Kerr Hypothesis*

When a perturbed black hole relaxes to equilibrium, it emits gravitational radiation in a characteristic superposition of damped sinusoids QNMs (Press 1971; Teukolsky 1973). In general relativity, the complex frequencies of these modes. $f_{\ell mn}$ for the oscillation frequency and $\tau_{\ell mn}$ for the damping time depend only on the mass and spin of the final black hole, not on the details of the perturbation that excited them. This follows directly from the no-hair theorem (Israel 1967) and forms the basis of black hole spectroscopy: each mode provides an independent measurement of the remnant's properties, and consistency among multiple modes constitutes a test of the Kerr metric.

The post-merger signal can be described as

$$h_+ - ih_\times = \sum_{\ell\geq 2, m\leq \ell, n} e^{-t/\tau_{\ell mn}} \left(A^R_{\ell mn} e^{-2\pi i f_{\ell mn} t} + A^L_{\ell mn} e^{2\pi i f_{\ell mn} t}\right) \tag{1}$$

where the complex amplitudes $A^{R,L}_{\ell mn}$ encode the excitation of each mode by the progenitor dynamics.

3.2. *Evidence for Multiple Modes in GW250114*

Previous ringdown analyses of gravitational-wave events have often been limited to detecting only the fundamental quadrupolar mode (220). GW250114, with its exceptional SNR, has changed this fact (Abac et al. 2025). Three independent analysis pipelines, RINGDOWN (Isi et al. 2019; Abac et al. 2025), pyRing (Carullo et al. 2019; Abac et al. 2025), and the QNM rational filter (QNMRF) (Ma et al. 2022; Abac et al. 2025) have been employed to study the post-merger signal.

The agnostic two-damped-sinusoid (2DS) model, which makes no assumptions about the mode frequencies, reveals that a second, more rapidly decaying component is required by the data. The amplitude of this second component is bounded away from zero at $> 3\sigma$ significance for fit start times up to $9t_{M_f}$ (Abac et al. 2025), where $t_{M_f} = (1+z)GM_f/c^3 \approx 0.337$ ms is the characteristic ringdown timescale. The measured frequencies and damping times of the two components are broadly consistent with the 220 (fundamental) and 221 (first overtone) modes predicted from the remnant mass and spin inferred from the full inspiral-merger-ringdown analysis (Abac et al. 2025) (These modes are shown in Fig. 2 red and green regions).

When fitting explicitly with a $220+221$ model, the amplitude of the 221 mode follows the expected exponential decay for fit start times $t_{\rm start} \geq 6t_{M_f}$ (Abac et al. 2025). At earlier times, deviations suggest the presence of additional content, such as the 222 overtone, though its amplitude is not confidently measured. The QNMRF analysis provides independent confirmation, finding strong support for the $220+221$ model over a single-mode model for $t_{\rm start} \leq 10t_{M_f}$, with weaker support for including the 222 mode at very early times (Abac et al. 2025).

3.3. *Tests of the Kerr Nature*

The spectroscopic pattern of multiple modes provides a direct test of whether the remnant is a Kerr black hole. By allowing the frequency and damping time of the 221 mode to vary freely from their Kerr-predicted values, parameterized as

$$\delta\hat{f}_{221} = \ln(f_{221}/f_{221,\rm Kerr}), \quad \delta\hat{\tau}_{221} = \ln(\tau_{221}/\tau_{221,\rm Kerr}) \tag{2}$$

the analysis constrains these deviations to $\delta\hat{f}_{221} = -0.13^{+0.16}_{-0.16}$ at $t_{\rm start} = 3t_{M_f}$ and $\delta\hat{f}_{221} = -0.07^{+0.72}_{-0.53}$ at $t_{\rm start} = 9t_{M_f}$ (90% credible intervals) (Abac et al. 2025). These measurements are fully consistent with the Kerr prediction ($\delta\hat{f} = \delta\hat{\tau} = 0$) and represent the most precise single-event constraints on overtone frequencies to date (Fig. 2 shows the 90% credible regions for these modes alongside the Kerr predictions.).

Furthermore, the relative amplitudes and phases of the 220 and 221 modes show consistency at the $\geq 38\%$ credible level with a numerical relativity simulation of a binary with parameters close to GW250114, spanning fit start times from $3t_{M_f}$ to $9t_{M_f}$ (Abac et al. 2025). This agreement between observation and simulation in the strong-field regime provides additional validation of general relativity.

Previous efforts to detect multiple ringdown modes have yielded mixed results, with some studies claiming evidence for overtones in GW150914 and others questioning those claims. The hierarchical combination of 21 events from GWTC-3 constrained the 221 mode deviation to $\delta\hat{f}_{221} = 0.01^{+0.27}_{-0.28}$ (Abbott et al. 2025). GW250114 alone achieves comparable precision, demonstrating the power of high-SNR events for black hole spectroscopy (Abac et al. 2025).

4. FULL INSPIRAL-MERGER-RINGDOWN TESTS

While isolating the ringdown phase provides clean tests of the remnant, incorporating the full signal, including inspiral and merger enables more stringent constraints by leveraging the entire information content of the waveform (Brito et al. 2018). The pSEOBNR analysis introduces fractional deviations to the frequency and damping time of fundamental QNMs within the SEOBNRv5PHM waveform model (Pompili et al. 2023):

$$f_{\ell m 0} = f^{\rm GR}_{\ell m 0}(1+\delta\hat{f}_{\ell m 0}), \quad \tau_{\ell m 0} = \tau^{\rm GR}_{\ell m 0}(1+\delta\hat{\tau}_{\ell m 0}) \tag{3}$$

This approach assumes GR holds through the merger, with potential deviations only affecting the ringdown frequencies, and uses the full IMR signal to constrain these parameters (Abac et al. 2025).

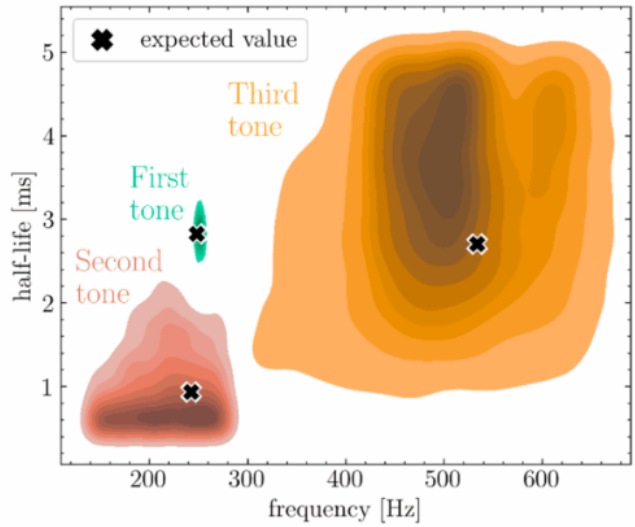

Figure 2. Constraints on quasinormal mode frequencies and damping times for the GW250114 remnant. The color denotes probability regions, from 90% (lightest) to 10% (darkest). The analysis shows 90% credible intervals from different analyses: the 221 overtone from ringdown-only analysis (red), the 220 fundamental mode from full-signal pSEOBNR analysis (green), and the 440 higher-mode (yellow). Black markers indicate Kerr predictions based on the remnant mass and spin inferred from the full inspiral-merger-ringdown analysis. All measurements are consistent with the Kerr metric. From LIGO, Virgo and KAGRA (LVK) collaborations.

4.1. *Constraints on the 220 and 440 Modes*

For GW250114, the pSEOBNR analysis yields exceptionally tight constraints on the fundamental quadrupolar mode: $\delta\hat{f}_{220} = 0.02^{+0.02}_{-0.02}$ and $\delta\hat{\tau}_{220} = -0.01^{+0.10}_{-0.09}$ (90% credible intervals) (Abac et al. 2025). These are approximately twice as stringent as the constraints obtained by hierarchically combining 17 events from GWTC-4.0, which gave $\delta\hat{f}_{220} = 0.00^{+0.06}_{-0.06}$ and $\delta\hat{\tau}_{220} = 0.16^{+0.18}_{-0.16}$ (Abac et al. 2025). The GR quantile $\mathcal{Q}_{\rm GR} = 54.2\%$ indicates excellent consistency with Einstein's theory (Abac et al. 2025).

The high SNR of GW250114 also enables constraints on higher-order modes for the first time in a single event.

The hexadecapolar (4,4) fundamental mode, which contributes SNR $3.6^{+1.4}_{-1.5}$ to the signal, is constrained to $\delta\hat{f}_{440} = -0.12^{+0.36}_{-0.34}$ (Abac et al. 2025). While this constraint is not yet as tight as that on the 220 mode, it represents the first measurement of a non-quadrupolar ringdown mode in an individual event and opens the door to multimode spectroscopy with higher harmonics. The reconstructed complex frequencies yield $f_{220} = 251.7^{+5.1}_{-5.0}$ Hz, $\tau_{220} = 4.09^{+0.42}_{-0.38}$ ms, $f_{440} = 503^{+130}_{-185}$ Hz, and $\tau_{440} = 4.7^{+2.1}_{-2.7}$ ms (Abac et al. 2025) (The 440 mode constraints are shown as the yellow region in Fig. 2). Inverting these frequencies to estimate the remnant's mass and spin gives results consistent with both the IMR-inferred values and the ringdown-only analyses, providing a cross-consistency check across different portions of the signal.

4.2. *Implications for Modified Gravity*

These stringent constraints can be mapped to specific alternative gravity theories. As an example, consider dynamical Chern-Simons (dCS) gravity, a parity-violating extension of GR in which the black hole ringdown spectrum receives corrections controlled by a coupling length $\sqrt{\alpha_{\rm dCS}}$ (Touchette 2009). Using the bound on the 220 mode frequency, one obtains an approximate constraint $\sqrt{\alpha_{\rm dCS}} < 32.2$ km for purely axial perturbations (or < 40.1 km for polar perturbations) (Abac et al. 2025). This single-event bound is competitive with recent ringdown-only analyses of GW150914, GW190521, and GW200129 (Chung et al. 2025), demonstrating the power of high-SNR events to constrain beyond-GR theories (Abac et al. 2025).

The consistency of multiple modes with the Kerr prediction also places strong constraints on exotic compact objects, which would exhibit different ringdown spectra due to modified boundary conditions at the horizon scale (Cardoso et al. 2016).

5. INSPIRAL PHASE

The inspiral phase of binary black hole mergers is well described by post-Newtonian (PN) theory, an expansion in powers of v/c (Blanchet 2002). At each PN order, the gravitational-wave phase accumulates a contribution characterized by coefficients ϕ_i. Parameterized tests of GR introduce fractional deviations $\delta\hat{\phi}_i$ at each order, allowing the data to constrain possible departures from Einstein's theory (Yunes & Pretorius 2009).

Two independent pipelines are employed for these tests: the Flexible Theory Independent (FTI) approach (Mehta et al. 2023) and the Test Infrastructure for General Relativity (TIGER) (Agathos et al. 2014). These pipelines differ in their waveform models, cutoff frequencies, and technical implementations, providing complementary cross-checks. The FTI pipeline employs waveform models calibrated to numerical relativity (Mehta et al. 2023), while TIGER utilizes the precessing-spin $IMRPhenomXPHM_{SpinTaylor}$ model (Pratten et al. 2021).

5.1. *Constraints from GW250114*

Due to its exceptional SNR during the inspiral phase, GW250114 provides remarkably tight constraints on PN coefficients. The leading-order (0PN) coefficient is constrained to $\delta\hat{\phi}_0 = 0.00^{+0.03}_{-0.03}$, while the 1.5PN coefficient (which includes spin-orbit effects) is constrained to $\delta\hat{\phi}_3 = -0.01^{+0.03}_{-0.02}$ (90% credible intervals) (Abac et al. 2025). These are 2–3 times more stringent than the constraints obtained by hierarchically combining all GWTC-4.0 events, which gave $\delta\hat{\phi}_0 = -0.00^{+0.09}_{-0.09}$ and $\delta\hat{\phi}_3 = 0.00^{+0.07}_{-0.07}$ (Abac et al. 2025).

Fig. 3 illustrates this dramatic improvement, showing the GW250114 constraints (blue stars) alongside individual GWTC-4.0 events (light red bands) and the hierarchical catalog combination (red squares). Across all post-Newtonian orders from -1PN to 3.5PN, GW250114 achieves bounds 2-3 times tighter than the full catalog.

Notably, GW250114 does not place competitive constraints on the -1PN coefficient (which would correspond to dipole radiation in alternative theories) compared to GW170817, as the binary neutron star merger remains the most sensitive probe of dipole emission due to its longer inspiral duration (Abbott et al. 2019).

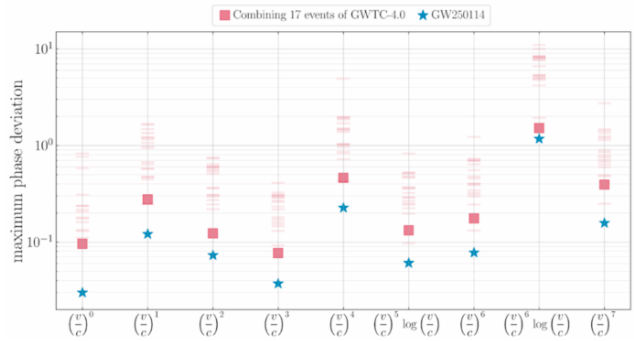

Figure 3. 90% upper bounds on post-Newtonian deviation parameters from GW250114 (blue stars) compared to individual GWTC-4.0 events (light red bands) and their hierarchical combination (red squares). The constraints shown combine results from multiple analysis pipelines. GW250114 achieves bounds 2-3 times more stringent than the full catalog across all PN orders. Image from LIGO, Virgo and KAGRA (LVK) collaborations.

Individual PN coefficients are correlated in their effects

on the waveform. Principal component analysis (PCA) identifies the linear combinations of coefficients that are best constrained by the data (Saleem et al. 2022). For GW250114, the leading PCA component (which captures the dominant mode of deviation across the 1.5PN to 3.5PN range) is constrained to $\delta\hat{\phi}^{(1)}_{\rm PCA} = -0.01^{+0.02}_{-0.02}$, slightly better than the best-measured individual coefficient (Abac et al. 2025). The subleading component is constrained to $\delta\hat{\phi}^{(2)}_{\rm PCA} = 0.04^{+0.14}_{-0.15}$ (Abac et al. 2025). Both are fully consistent with GR.

6. SIGNAL CONSISTENCY TESTS

A powerful test of GR involves checking whether different portions of the signal yield consistent estimates of the remnant black hole properties (Ghosh et al. 2016). The signal is split at a cutoff frequency $f_c^{\rm IMR}$ (taken as the innermost stable circular orbit frequency of the remnant) (Bardeen et al. 1972), and the remnant mass and spin are inferred independently from the low-frequency (inspiral-dominated) and high-frequency (merger-ringdown-dominated) segments.
The deviation parameters are defined as

$$\frac{\Delta M_f}{\bar{M}_f} = \frac{2(M_f^{\rm insp} - M_f^{\rm postinsp})}{M_f^{\rm insp} + M_f^{\rm postinsp}}, \quad \frac{\Delta\chi_f}{\bar{\chi}_f} = \frac{2(\chi_f^{\rm insp} - \chi_f^{\rm postinsp})}{\chi_f^{\rm insp} + \chi_f^{\rm postinsp}} \tag{4}$$

with the GR expectation being zero for both. For GW250114, we obtain $\Delta M_f/\bar{M}_f = 0.02^{+0.07}_{-0.06}$ and $\Delta\chi_f/\bar{\chi}_f = -0.01^{+0.11}_{-0.11}$ (90% credible intervals) (Abac et al. 2025). These constraints are comparable in stringency to the hierarchical combination of all 12 events first reported in GWTC-4.0, which yielded $\Delta M_f/\bar{M}_f = 0.03^{+0.13}_{-0.11}$ and $\Delta\chi_f/\bar{\chi}_f = -0.01^{+0.13}_{-0.13}$ (Abac et al. 2025). The two-dimensional GR quantile for GW250114 is 49.5%, indicating excellent consistency (Abac et al. 2025).

6.1. *Test of the Hawking Area Theorem*

The second law of black hole thermodynamics, embodied in the Hawking area theorem, states that the total horizon area of black holes cannot decrease (Hawking 1971). For a binary black hole merger, this implies that the area of the final remnant must exceed the sum of the areas of the two progenitors. Using the inspiral and post-inspiral parameter estimates, one can compute the initial and final horizon areas and assess the statistical significance of any violation.
For GW250114, this test yields a credibility of 4.8σ (Abac et al. 2025) in favor of the area theorem, meaning the data favor an increase in horizon area at this statistical significance. Figure 4 shows the probability distribution for the fractional area change $\Delta\mathcal{A}/\mathcal{A}_i = (\mathcal{A}_f - \mathcal{A}_i)/\mathcal{A}_i$. The distribution strongly disfavors negative values (left of the dashed line at zero) and peaks near the General Relativity prediction of approximately 65% area increase (light gray band). Represents one of the strongest observational confirmations of the second law of black hole mechanics to date.

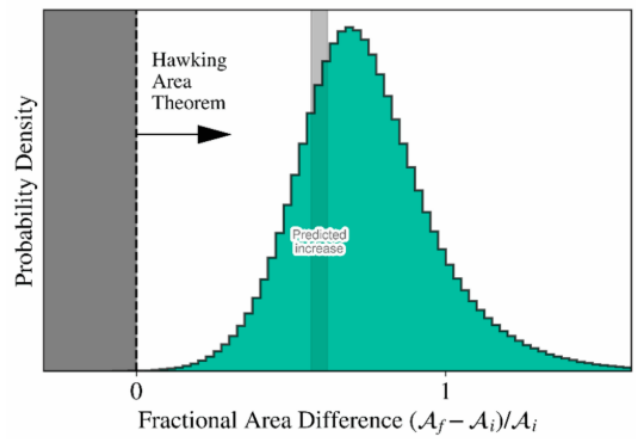

Figure 4. Probability distribution for the fractional area change $\Delta\mathcal{A}/\mathcal{A}_i$ in the GW250114 merger. Values greater than zero (right of dashed line) indicate an increase in total horizon area, consistent with the Hawking area theorem. The distribution peaks near the General Relativity prediction of approximately 65% area increase (light gray band) and disfavors area decrease at 4.8σ significance. From LIGO, Virgo and KAGRA (LVK) collaborations.

6.2. *Residuals Test*

A final consistency check searches for any excess coherent power in the detector network after subtracting the best-fit waveform (Cornish et al. 2011). Such residuals could indicate deviations from GR, unmodeled physics, or instrumental artifacts. For GW250114, the upper limit on the residual network SNR is 6.86 at 90% credibility, with a p-value of 0.34 (Abac et al. 2025). No statistically significant excess power is found, indicating that the signal is well described by the GR waveform model (Abac et al. 2025).

7. CONCLUSION

With a network SNR of 76, this single event has enabled the most stringent tests of general relativity and the Kerr nature of black holes ever performed. At least three quasinormal modes have been identified or constrained, the fundamental quadrupolar mode (220), its first overtone (221), and the hexadecapolar mode (440), and their spectroscopic pattern aligns perfectly with the Kerr metric prediction. The amplitudes and phases of these modes match numerical relativity simulations of

comparable systems in GR.

Across the inspiral, merger, and ringdown, constraints from GW250114 alone are comparable to, and in some cases 2-3 times more stringent than, those obtained by hierarchically combining dozens of events from the GWTC-4.0 catalog. The signal passes all consistency tests, including checks of the Hawking area theorem at 4.8σ credibility and residuals analyses showing no excess power beyond noise. These results also place competitive constraints on alternative gravity theories, such as dynamical Chern-Simons gravity, and rule out large classes of exotic compact objects.

A decade after the first detection of gravitational waves, general relativity and the Kerr metric remain empirically unshaken.

8. ACKNOWLEDGMENT

Almost all numerical results are obtained from paper Abac et al. (2025).

REFERENCES

Abac, A. G., Abouelfettouh, I., Acernese, F., et al. 2025, Phys. Rev. Lett., 135, 111403, doi: 10.1103/kw5g-d732

Abbott, B. P., Abbott, R., Abbott, T. D., et al. 2016, Phys. Rev. Lett., 116, 061102, doi: 10.1103/PhysRevLett.116.061102

—. 2019, Phys. Rev. Lett., 123, 011102, doi: 10.1103/PhysRevLett.123.011102

Abbott, R., Abe, H., Acernese, F., et al. 2025, Phys. Rev. D, 112, 084080, doi: 10.1103/PhysRevD.112.084080

Agathos, M., Del Pozzo, W., Li, T. G. F., et al. 2014, Phys. Rev. D, 89, 082001, doi: 10.1103/PhysRevD.89.082001

Almheiri, A., Hartman, T., Maldacena, J., Shaghoulian, E., & Tajdini, A. 2021, Rev. Mod. Phys., 93, 035002, doi: 10.1103/RevModPhys.93.035002

Bagui, E., Clesse, S., Luca, V. D., et al. 2023, Primordial black holes and their gravitational-wave signatures. https://arxiv.org/abs/2310.19857

Bardeen, J. M., Press, W. H., & Teukolsky, S. A. 1972, ApJ, 178, 347, doi: 10.1086/151796

Blanchet, L. 2002, Living Reviews in Relativity, 5, doi: 10.12942/lrr-2002-3

Brito, R., Buonanno, A., & Raymond, V. 2018, Phys. Rev. D, 98, 084038, doi: 10.1103/PhysRevD.98.084038

Cardoso, V., Hopper, S., Macedo, C. F. B., Palenzuela, C., & Pani, P. 2016, Phys. Rev. D, 94, 084031, doi: 10.1103/PhysRevD.94.084031

Carullo, G., Del Pozzo, W., & Veitch, J. 2019, Phys. Rev. D, 99, 123029, doi: 10.1103/PhysRevD.99.123029

Chung, A. K.-W., Lam, K. K.-H., & Yunes, N. 2025, Phys. Rev. D, 111, 124052, doi: 10.1103/g83n-rrlj

Clifton, T., Ferreira, P. G., Padilla, A., & Skordis, C. 2012, Physics Reports, 513, 1, doi: https://doi.org/10.1016/j.physrep.2012.01.001

Collaboration, T. L. S., Aasi, J., Abbott, B. P., et al. 2015, Classical and Quantum Gravity, 32, 074001, doi: 10.1088/0264-9381/32/7/074001

Cornish, N., Sampson, L., Yunes, N., & Pretorius, F. 2011, Phys. Rev. D, 84, 062003, doi: 10.1103/PhysRevD.84.062003

Detweiler, S. 1980, ApJ, 239, 292, doi: 10.1086/158109

Einstein, A. 1915, Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften, 778

Ghosh, A., Ghosh, A., Johnson-McDaniel, N. K., et al. 2016, Phys. Rev. D, 94, 021101, doi: 10.1103/PhysRevD.94.021101

Gossan, S., Veitch, J., & Sathyaprakash, B. S. 2012, Phys. Rev. D, 85, 124056, doi: 10.1103/PhysRevD.85.124056

Hawking, S. W. 1971, Phys. Rev. Lett., 26, 1344, doi: 10.1103/PhysRevLett.26.1344

Hawking, S. W., & Penrose, R. 1970, Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences, 314, 529, doi: 10.1098/rspa.1970.0021

Isi, M., Giesler, M., Farr, W. M., Scheel, M. A., & Teukolsky, S. A. 2019, Phys. Rev. Lett., 123, 111102, doi: 10.1103/PhysRevLett.123.111102

Israel, W. 1967, Phys. Rev., 164, 1776, doi: 10.1103/PhysRev.164.1776

Johnson, H. M. 1981, ApJ, 243, 234, doi: 10.1086/158589

Kerr, R. P. 1963, Phys. Rev. Lett., 11, 237, doi: 10.1103/PhysRevLett.11.237

Kramer, M., Stairs, I. H., Manchester, R. N., et al. 2021, Phys. Rev. X, 11, 041050, doi: 10.1103/PhysRevX.11.041050

Ma, S., Mitman, K., Sun, L., et al. 2022, Phys. Rev. D, 106, 084036, doi: 10.1103/PhysRevD.106.084036

Mehta, A. K., Buonanno, A., Cotesta, R., et al. 2023, Phys. Rev. D, 107, 044020, doi: 10.1103/PhysRevD.107.044020

Pompili, L., Buonanno, A., Estellés, H., et al. 2023, Phys. Rev. D, 108, 124035, doi: 10.1103/PhysRevD.108.124035

Pratten, G., García-Quirós, C., Colleoni, M., et al. 2021, Phys. Rev. D, 103, 104056, doi: 10.1103/PhysRevD.103.104056

Press, W. H. 1971, ApJL, 170, L105, doi: 10.1086/180849

Raju, S. 2022, Physics Reports, 943, 1, doi: https://doi.org/10.1016/j.physrep.2021.10.001

Saleem, M., Datta, S., Arun, K. G., & Sathyaprakash, B. S. 2022, Phys. Rev. D, 105, 084062, doi: 10.1103/PhysRevD.105.084062

Teukolsky, S. A. 1973, ApJ, 185, 635, doi: 10.1086/152444

Touchette, H. 2009, Physics Reports, 478, 1, doi: https://doi.org/10.1016/j.physrep.2009.05.002

Varma, V., Field, S. E., Scheel, M. A., et al. 2019, Phys. Rev. Res., 1, 033015, doi: 10.1103/PhysRevResearch.1.033015

Visser, M., & Wiltshire, D. L. 2004, Classical and Quantum Gravity, 21, 1135, doi: 10.1088/0264-9381/21/4/027

Will, C. M. 2014, Living Reviews in Relativity, 17, doi: 10.12942/lrr-2014-4

Yunes, N., & Pretorius, F. 2009, Phys. Rev. D, 80, 122003, doi: 10.1103/PhysRevD.80.122003